

Warm-up 1

Compute the following matrix powers:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^8 =$$

$$\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}^8 =$$

During class 1

Give an example of a real 2×2 matrix for each of the four regions of a Venn diagram where the two categories are “invertible” and “diagonalizable”; or for a given region, explain why no such matrix exists.

How would your answer change if we specified complex 2×2 matrices instead?